Zero- Versus Infinite-Temperature Damping in Variational Quantum Circuits

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Randomized compiling turns amplitude damping (AD) into its Pauli twirl, which keeps the contraction of the damping but removes its non-unital term. The twirl is generalized amplitude damping at infinite temperature, so comparing the two isolates the zero-temperature bias of the damping. A recent study reported that twirling degrades variational models. It remains unclear whether this reflects a real loss in what the models can learn or only a change in how their outputs must be read out. In this work, we compare AD with its Pauli and Clifford twirls in exact simulations of a re-uploading classifier. We show that the bias acts mainly through the spread of the measured expectation values over the inputs. Under AD, this spread settles on a floor that vanishes at infinite temperature, whereas under the twirls it keeps shrinking with every layer. A trainable readout recovers the noiseless accuracy under every channel. The difference reappears instead as a cost in measurement shots. At a fixed shot budget, the classifier under AD stays close to its noiseless accuracy, while the twirled classifiers fall far below it. At weak damping, the two separate only in very deep circuits.

I. INTRODUCTION

Variational quantum algorithms and quantum machine learning models are built from parameterized circuits and are meant to run on noisy near-term devices [1–3]. Whether such a model can be trained depends on how strongly its outputs vary with the parameters. In deep random circuits, this variation vanishes exponentially with the number of qubits, a phenomenon known as a barren plateau [4–6], and noise can make it worse. Unital noise drives the state toward the maximally mixed state, so local expectation values decay exponentially with depth [7, 8]. Non-unital noise behaves differently. Amplitude damping (AD) drives the state toward a pure state and avoids barren plateaus for local observables [9], although the output then forgets the early layers [10, 11].

This difference matters in practice, because hardware noise is often made unital on purpose. Randomized compiling [12, 13] averages the noise over random Pauli gates, which turns AD into its Pauli twirl. The twirl keeps the contraction of AD but removes its non-unital term, the pull toward $|0\rangle$, and leaves generalized amplitude damping at infinite temperature. Comparing AD with its Pauli twirl therefore isolates the zero-temperature bias of the damping. A recent study [14] made this comparison and found that twirling shrinks gradients and degrades the performance of variational models. What these effects mean is less clear. They could reflect a real loss in what the model can learn, or only a change in how its outputs must be read out.

In this work, we show that in our classifier the zero-temperature bias acts mainly through the spread of the measured expectation values over the inputs. Under AD, this spread settles on a floor, whereas under the twirls it keeps shrinking with every layer. The floor vanishes as the temperature rises. A trainable readout absorbs most of the resulting differences in accuracy, much as rescaling methods of error mitigation do [15], so the twirled models lose little in what they can learn. The difference reappears instead as a cost in measurement shots [16]: at a fixed shot budget, the classifier under AD stays close to its noiseless accuracy, while the twirled classifiers fall far below it. At weak damping, the two separate only in very deep circuits. Together, these findings explain the reported degradation as a loss of output scale, which a trained readout restores at a cost in shots.

We organize the text as follows. Section II introduces the channels and the classifier. Section III treats the output scale, and Sec. IV the spread, the sampling cost, and their dependence on the damping strength. Section V concludes.

II. CHANNELS AND MODEL

In this section, we introduce the noise channels that we compare and the four-qubit re-uploading classifier on which we compare them.

A. Channels

We describe each single-qubit channel $\mathcal{N}$ through its action on the Bloch vector, which is an affine map $\mathbf{r} \mapsto T\mathbf{r} + \mathbf{t}$ [17], with

$$T_{ij} = \tfrac{1}{2}\mathrm{Tr}\big[P_i\,\mathcal{N}(P_j)\big], \qquad t_i = \tfrac{1}{2}\mathrm{Tr}\big[P_i\,\mathcal{N}(I)\big], \tag{1}$$

where $P_i \in \{X, Y, Z\}$. The matrix T describes how the channel shrinks the Bloch vector, and the vector $\mathbf{t}$ how it pushes it. The channel is unital if and only if $\mathbf{t} = \mathbf{0}$.

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We compare amplitude damping with two twirled versions of it. Amplitude damping of strength p does two things to the Bloch vector. It shrinks the vector, by $c = \sqrt{1-p}$ in the transverse directions and by c^2 along z, and it pushes the vector toward $|0\rangle$ by p. The Clifford twirl [18] averages these effects over all directions and leaves depolarizing noise, which shrinks every component by the same factor and pushes nowhere. The Pauli twirl [12, 19] is gentler. It keeps the shrinking of AD exactly but removes the push toward $|0\rangle$, the non-unital term.

This comparison has a simple physical meaning. Generalized amplitude damping [17] with excited-state population N has the same T as AD and $t_z = p(1-2N)$. AD is the case $N = 0$, and its Pauli twirl is the case $N = \frac{1}{2}$. Comparing AD with its Pauli twirl therefore compares damping at zero and at infinite temperature.

We treat the twirled channels as idealized, replacing the channel by its twirl at every insertion. No twirling protocol can do this exactly after a non-Clifford gate such as the data encoding, so the twirled channels should be read as model channels that share the contraction of AD.

B. Four-qubit re-uploading classifier

Our model is a data re-uploading classifier [20, 21] on $n = 4$ qubits (Fig. 1), which classifies MNIST images [22] of the digits $\{1, 3, 5\}$. Each image is reduced to four principal components, which are mapped linearly to angles $a_j \in [0, 2\pi]$. As a second data set, we use three classes of Fashion-MNIST [23].

The circuit is built from three ingredients, namely trainable blocks, the encoding, and a ring of CNOTs,

$$\begin{aligned} \mathcal{R}^{(\ell)} = \bigotimes_{j=1}^{n} R_Z R_Y R_Z, \quad S(\mathbf{a}) = \bigotimes_{j=1}^{n} R_X(a_j), \\ E = \mathrm{CNOT}_{n,1} \cdots \mathrm{CNOT}_{2,3}\, \mathrm{CNOT}_{1,2}, \end{aligned} \tag{2}$$

where $\mathrm{CNOT}_{j,k}$ has control j and target k. Each layer applies a trainable block and the ring, then the encoding, with the channel acting after both,

$$\mathcal{L}_{\mathbf{a}}^{(\ell)} = \mathcal{N}^{\otimes n} \circ \mathcal{U}\big(S(\mathbf{a})\big) \circ \mathcal{N}^{\otimes n} \circ \mathcal{U}\big(E\,\mathcal{R}^{(\ell)}\big), \tag{3}$$

and the full circuit is

$$\Phi_{\mathbf{a},\theta} = \mathcal{N}^{\otimes n} \circ \mathcal{U}\big(\mathcal{R}^{(L)}\big) \circ \mathcal{L}_{\mathbf{a}}^{(L-1)} \circ \cdots \circ \mathcal{L}_{\mathbf{a}}^{(0)}, \tag{4}$$

with $2L+1$ insertions of the channel per qubit and θ the $3n(L+1)$ trainable angles of the blocks. The readout acts on the single-qubit expectation values

$$z_j(\mathbf{a}, \theta) = \mathrm{Tr}\big[Z_j\, \Phi_{\mathbf{a},\theta}\big(|0\rangle\langle 0|^{\otimes n}\big)\big], \tag{5}$$

and the scores of the three classes are

$$\ell(\mathbf{a}) = \begin{cases} W\mathbf{z} + \mathbf{b} & \text{(raw)}, \\ W\,\mathrm{diag}(\boldsymbol{\sigma})^{-1}(\mathbf{z} - \boldsymbol{\mu}) + \mathbf{b} & \text{(standardized)}, \end{cases} \tag{6}$$

where W is a trainable $3 \times n$ weight matrix and $\mathbf{b}$ a trainable bias. In the standardized readout, $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ collect the means μ_j and the population standard deviations σ_j of the z_j over the 1000 training images. We recompute them at every training step, with gradients flowing through them, and apply their values at the final parameters to the test images. The standardized readout adds no parameters to the raw one. It only fixes the scale at which W acts.

Unless stated otherwise, we use $p = 0.3$ and simulate the density matrix exactly, so the expectation values carry no sampling error. The exception is the finite-shot runs, in which each z_j carries the noise of N_s measurements, in training as well as at test time. For these runs, we define the operational shot cost of a trained model as the smallest N_s at which its mean test accuracy lies within 0.01 of its accuracy with exact expectation values, and we report its geometric mean over the initializations. Training settings, classical baselines, and the shot model are given in Appendix C.

III. OUTPUT SCALE

We begin with the simplest way in which noise can change a variational model: by changing the scale of its output. AD and its Pauli twirl, damping at zero and at infinite temperature, share the same contraction matrix T. Any difference in scale between them must therefore come from the non-unital vector $\mathbf{t}$, which carries the zero-temperature bias (Sec. IV). In this section, we ask how much of the effect of the noise a trainable output scale can undo, and what it costs to undo it.

A. Scale and sampling cost

Consider first the simplest case, in which the noise only rescales and shifts each expectation value,

$$\tilde{z}_j = s\, z_j + b_j, \tag{7}$$

with constants $s \neq 0$ and b_j that depend on neither the input nor the parameters. A trainable affine readout can then undo the noise completely: every classifier that it can build from the noiseless $\mathbf{z}$, it can also build from the noisy $\tilde{\mathbf{z}}$, so the noise leaves the model class unchanged. A single qubit under depolarizing noise is an exact example. Because this channel commutes with every unitary, N_{ins} insertions simply multiply its expectation value $f = \langle Z \rangle$ by a constant,

$$\tilde{f} = \lambda_{\mathrm{depol}}^{N_{\mathrm{ins}}} f, \tag{8}$$

where $\lambda_{\mathrm{depol}} = \frac{1}{3}(2c + c^2)$ is the factor by which depolarizing noise shrinks each Bloch component, the average of those of AD. For anisotropic or non-unital channels, or on more than one qubit, the noisy model is no longer a rescaled copy of the noiseless one. Whether the scale still

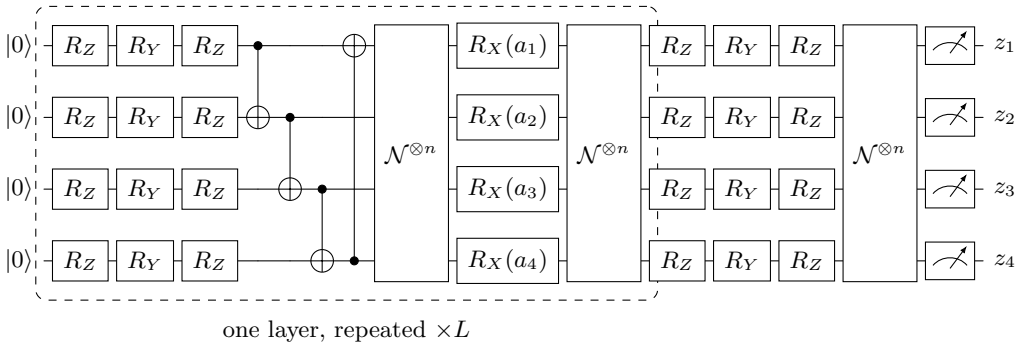


FIG. 1. The classifier at $n = 4$, drawn for one layer and the final block. Each $R_Z R_Y R_Z$ triple carries three trainable angles; the encoding column re-uploads the angle vector $\mathbf{a}$ in each of the L layers. The channel acts on every qubit after the entangler, after the encoding, and after the final block, giving $2L + 1$ insertions per qubit. The expectation values $z_j = \langle Z_j \rangle$ enter a trainable affine readout.

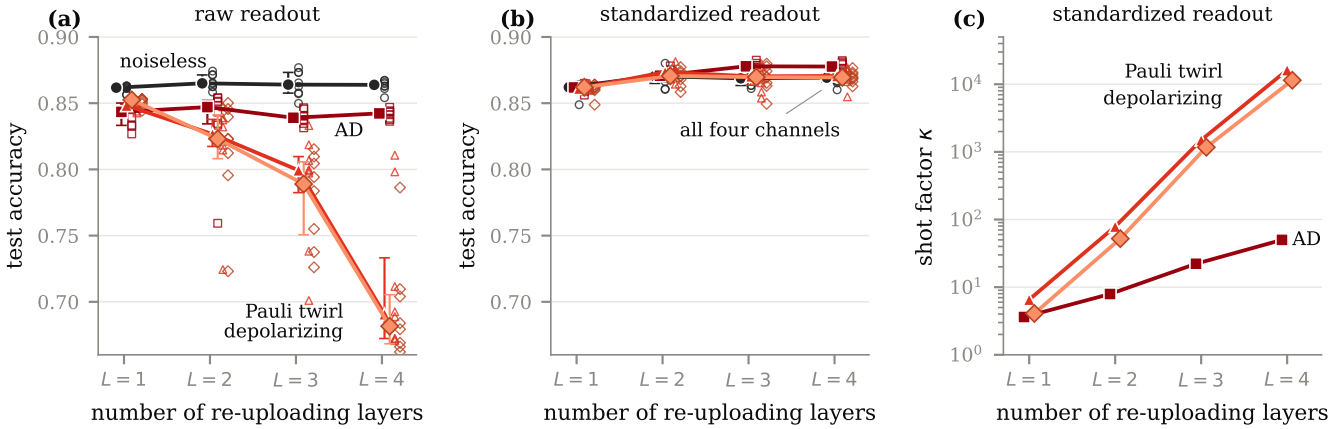


FIG. 2. Four-qubit classifier on MNIST against the number of re-uploading layers. (a) Test accuracy with the raw readout and (b) with the standardized readout, both after 100 steps, and (c) shot factor κ of Eq. (10) with the standardized readout. Panels (a) and (b) show the median and interquartile range over eight initializations, with hollow markers for the individual runs; one Pauli run at $L = 1$ in (b) diverged and lies below the axis.

accounts for most of the difference then becomes an empirical question, which we test in the rest of this section. The average behavior of the spread on several qubits, its per-layer decay under the twirls and its floor under AD, is derived in Secs. IV A and IV C.

On hardware, however, rescaling is not free. The number of shots is one of the main costs of variational algorithms [24, 25], and an expectation value estimated from N_s shots, $\hat{z}_j$, has the variance

$$\mathrm{Var}(\hat{z}_j) = \frac{1 - z_j^2}{N_s} \leq \frac{1}{N_s}. \tag{9}$$

The standardized readout divides each z_j by its spread σ_j, so the shot noise of the standardized z_j has variance $(1 - z_j^2)/(N_s\sigma_j^2) \leq 1/(N_s\sigma_j^2)$. The smaller the spread, the more the shot noise is amplified. Keeping the class scores equally precise therefore requires $N_s \propto 1/\sigma^2$, and a channel $\mathcal{N}$ needs about $\kappa(\mathcal{N})$ times the shots of the noiseless circuit, with the shot factor

$$\kappa(\mathcal{N}) = \left(\frac{\sigma(\mathrm{None})}{\sigma(\mathcal{N})} \right)^2, \tag{10}$$

where σ is the spread σ_j at the trained parameters, averaged over the qubits and the initializations. In the language of error mitigation, the trained readout performs a rescaling [15, 26]. Such a rescaling cannot undo an exponential concentration without an exponential number of shots [16]. In Sec. IV B, we test how well κ predicts the operational shot cost of Sec. II B, the number of shots that a trained classifier actually needs.

The same scale also accounts for the reported smaller gradient magnitudes. Under Pauli noise, the loss gradients carry the same attenuation factors as the output [14]. For a rescaled expectation value, $\partial(s z_j)/\partial\theta = s\,\partial z_j/\partial\theta$, so a trained output scale restores the size of the gradients. It does not restore their signal-to-noise ratio under shot noise, which it amplifies by the same factor, so estimating the gradients to a fixed precision still costs κ.

B. Classifier

We now test this picture in the four-qubit classifier, where the scale compounds over qubits and layers. With the raw readout and 100 training steps, the unital channels at first seem to lose trainability as the circuit gets deeper [Fig. 2(a) and Table IV in Appendix A]. Between $L = 1$ and $L = 4$, the test accuracy falls from 0.847 to 0.715 for the Pauli twirl and from 0.851 to 0.695 for depolarizing noise, and the training accuracy falls with it, to 0.700 and 0.672. AD stays at 0.84, and the noiseless circuit at 0.86. Taken at face value, this would suggest that the non-unital term keeps the deep circuit trainable.

A closer look shows that the drop comes from the readout, not from the circuit. If we train the same raw readout for 1000 steps, the unital channels recover to 0.837 and 0.826 at $L = 4$. Over these extra steps, the norm of their readout weights grows from about 19 to 106 and 109, while for the noiseless circuit it grows only from 8 to 20. The reason is that, after 100 steps, the unital channels have shrunk the spread to about 5 and 4 percent of the noiseless spread, and to below 1 percent before training. With a spread this small, the raw readout cannot grow its weights fast enough. The circuit itself remains trainable. With the standardized readout, every channel reaches the accuracy of the noiseless circuit at every depth [Fig. 2(b)], and the Pauli and depolarizing channels are statistically indistinguishable from it (Table III).

We see the same behavior on Fashion-MNIST at $L = 4$. With the raw readout, the unital channels fall 16 and 17 points below the noiseless circuit (0.765), and AD falls 5 points below it. With the standardized readout, all three channels come within two points of the noiseless circuit [Fig. 6 in Appendix A], while their shot factors are 34 for AD and 9.9×10^3 and 8.2×10^3 for the Pauli and depolarizing channels.

The depolarizing channel at $L = 1$ gives an exact check of this picture. Each z_j then crosses all three noise insertions as a weight-one Pauli string, so the channel multiplies every z_j by the same factor λ_{depol}^3, whatever the parameters. Since the standardized readout removes a uniform scale, the depolarizing and noiseless classifiers should make identical predictions, and indeed they do, on every test image. AD is not a pure rescaling, yet with the standardized readout it too stays within one percentage point of the noiseless accuracy at every depth (Table III).

What the standardized readout cannot remove is the difference in spread [Fig. 2(c)]. At the end of training, the shot factor of the unital channels grows by a factor of about 14 per layer, from 4.1 at $L = 1$ to 1.1×10^4 at $L = 4$ for depolarizing noise and to 1.6×10^4 for the Pauli twirl. For AD, it grows by only about 2.4 per layer, from 3.6 to 50. Note that the shot factor is not a property of the channel alone. It also depends on what the model is trained for. With the raw readout, the optimizer moves the parameters toward less contracted expectation values, and after 1000 steps at $L = 4$ the shot factors are 11 for AD, 320 for the Pauli twirl, and 470 for depolarizing noise.

In summary, the zero-temperature bias acts in our classifier mainly through the spread of the expectation values, and the apparent loss of trainability under the unital channels is a loss of scale. A trained readout absorbs this loss, but at a cost in shots that grows with depth.

IV. WHAT THE NON-UNITAL TERM CHANGES

Section III showed that a trainable scale removes most of the difference in accuracy between AD and its twirls. AD and its Pauli twirl share the contraction matrix T, so what remains is the non-unital term, of length $\|\mathbf{t}\| = p$. In this section, we show what it changes: the spread of the expectation values before training, the number of shots needed after training, and how both depend on the damping strength.

A. Floor of the spread

We start before training, where the effect is cleanest. Here we measure the spread as

$$\sigma^2 = \mathbb{E}_\theta \frac{1}{n} \sum_{j=1}^{n} \mathrm{Var}_{\mathbf{a}}\, z_j, \tag{11}$$

the variance of the z_j over the inputs, averaged over the qubits and over random parameters, and we write σ_{AD}, σ_{Pauli}, and σ_{None} for its values under AD, the Pauli twirl, and no noise. At $p = 0.3$, the spread separates the channels with both depth and width (Fig. 3). Under the unital channels, the spread shrinks by a nearly constant factor per layer at every width. Under AD, in contrast, it settles on a floor after two or three layers. The ratio $\sigma_{\text{AD}}/\sigma_{\text{Pauli}}$ therefore grows geometrically, by a factor of about 3.3 per layer at $n = 4$, and its square, which is the ratio of their shot factors, grows by about 11 per layer. As the circuit gets wider, the noiseless spread falls, as expected for a scrambling circuit, but the floor barely moves. At $n = 8$ and $L \geq 4$, the AD spread is therefore as large as the noiseless one.

The floor comes from the non-unital term. To see this, consider a Z component propagated backward from the readout. Under AD it becomes $c^2 Z + t_z I$, with $t_z = p$, whereas the Pauli twirl keeps only $c^2 Z$. The identity part behaves very differently from the rest. Its expectation value is 1 in every state, so it reaches the readout no matter what the earlier layers do. The twirl has no such part, and every component keeps shrinking from layer to layer. Under AD, the damping after the last entangler branches the Z component of each z_j into the identity and adds

$$\delta z_j = p(1-p)\left[(1-p)\, v_z \cos a_j - c\, v_y \sin a_j\right] \tag{12}$$

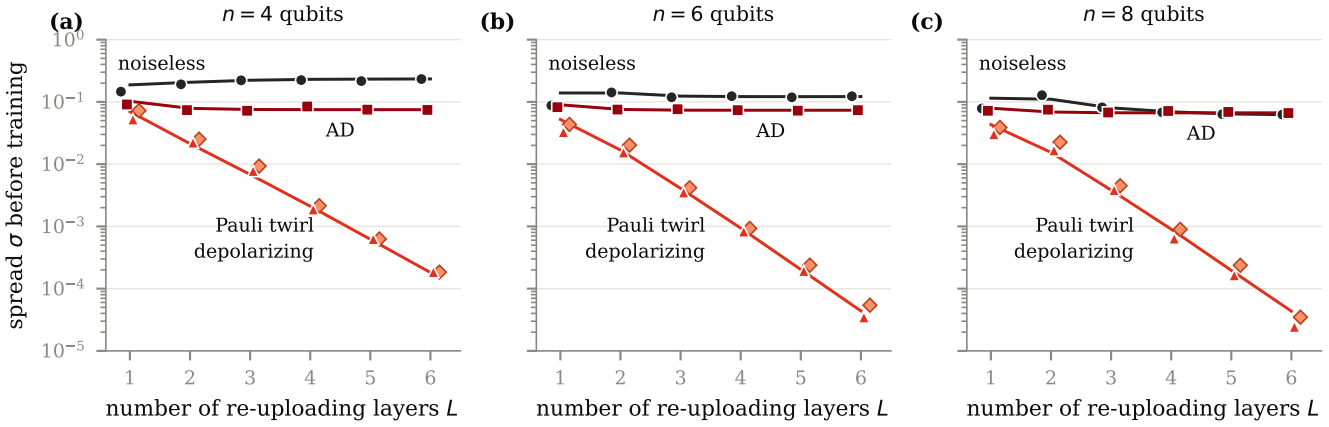


FIG. 3. Spread σ of the expectation values before training against the number of re-uploading layers at $p = 0.3$, for (a) four, (b) six, and (c) eight qubits, with n principal components as inputs. Markers are the root-mean-square spread over 300 training images and five parameter draws, shared across depths, for all four channels. Lines are the exact average over parameters for the noiseless circuit, AD, and the Pauli twirl (Appendix B).

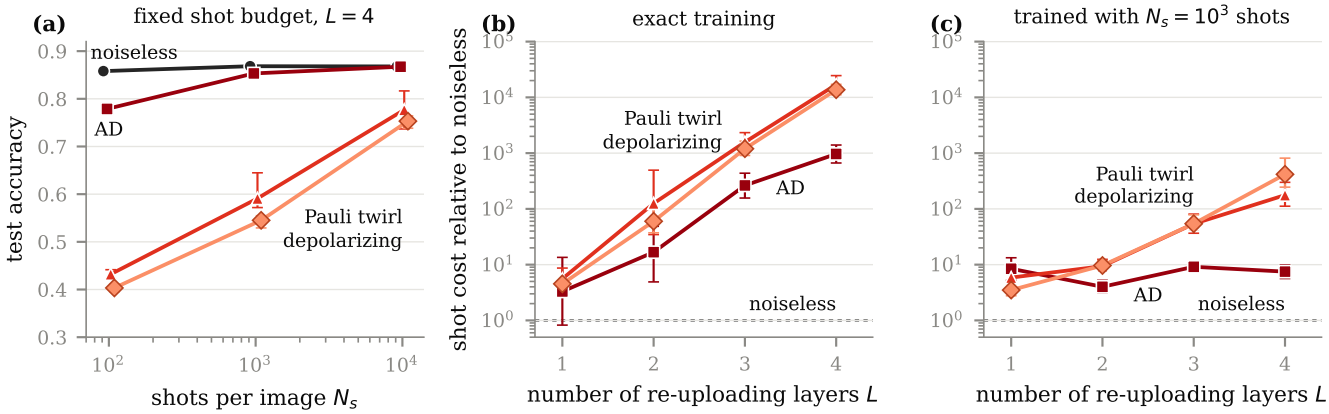


FIG. 4. Finite shots in the MNIST classifier at $p = 0.3$. (a) Test accuracy at $L = 4$ of models trained and tested with N_s shots per image (median and interquartile range over eight initializations). (b), (c) Operational shot cost relative to the noiseless circuit (geometric mean over initializations, with 95% bootstrap intervals over initializations) for models trained with exact expectation values (b) and with $N_s = 10^3$ shots (c).

to z_j, where (v_x, v_y, v_z) is the direction into which the final trainable block rotates Z_j. Averaging over the parameters, we obtain a closed form for this last-layer contribution,

$$\sigma_\infty^2 = \frac{p^2(1-p)^2}{n} \sum_j \Big[\frac{(1-p)^2}{2} \operatorname{Var}(\cos a_j) + \frac{1-p}{4} \operatorname{Var}(\sin a_j) \Big], \quad (13)$$

which at $p = 0.3$ matches the floor of the AD spread to within 0.5% for $n \geq 4$.

This floor fits well with what is known about noise-induced concentration. Unital noise concentrates local expectation values exponentially in depth [7, 10]. Under non-unital noise, by contrast, the variance of local expectation values over random circuits stays bounded from below by an amount set by the non-unital term [9]. Equation (13) is the counterpart of that bound for the variance over inputs. After the average over parameters, branchings in different layers are uncorrelated, so $\sigma_{\mathrm{AD}}^2 \geq \sigma_\infty^2$ at every depth. Both bounds scale as $\|\mathbf{t}\|^2 = p^2$ and do not decay with depth. At weak damping, the branchings of earlier layers raise the floor above this bound, as we show in Sec. IV C. At $p = 0.3$, however, the last layer alone sets the spread under AD. This is in line with the effective shallowness of noisy circuits [9]: only the last few layers before the readout shape the output.

The floor also shows how the zero-temperature bias depends on temperature. At finite temperature, the damping of a qubit with splitting $\hbar\omega$ relaxes toward the thermal excited-state population $N = (e^{\hbar\omega/k_B T} + 1)^{-1}$, and

Sec. II A gives

$$t_z = p(1-2N) = p\tanh\frac{\hbar\omega}{2k_BT} \tag{14}$$

at fixed p, that is, at fixed measured T_1. At fixed coupling to the bath, $1/T_1$ itself grows as $\coth(\hbar\omega/2k_BT)$. At weak damping, t_z is then independent of temperature to first order in p, and the temperature acts through the contraction instead. At fixed T_1, the picture is simple. The last-layer term, Eq. (12), is linear in t_z, so it scales exactly as $1-2N$. Strings of higher Pauli weight can branch into the identity on several qubits and add higher powers of t_z, but in the exact average over parameters they change this scaling by less than half a percent. The floor therefore vanishes linearly in $1-2N$ as the temperature rises, and it reaches zero at the Pauli twirl, $N=\frac{1}{2}$.

B. Sampling cost

We next test whether the shot factor predicts the number of shots that a trained classifier actually needs, by comparing it with the operational shot cost of Sec. II B. After training with exact expectation values, this cost follows κ for the unital channels but not for AD [Fig. 4(b) and Table II]. At $L=4$, AD needs 19 times more shots than its shot factor predicts. Even so, the Pauli twirl still needs 18 times the shots of AD. The extra cost of AD comes from the readout. The shot factor only looks at the average spread of the z_j, whereas the readout amplifies the shot noise of each z_j by its own weight. Exact training lets the AD readout rely on expectation values whose spread is far smaller than in the noiseless circuit. We capture this effect with the readout-weighted noise gain $G = \langle\sum_j \|\mathbf{w}_j\|^2(1-z_j^2)/\sigma_j^2\rangle$, where $\mathbf{w}_j$ is the column of W that multiplies z_j and the average runs over the test images. If the shot noise of different qubits is uncorrelated, G is N_s times the total shot-noise variance of the class scores. It reproduces the measured cost ratios to within a factor of 1.6 for $L\geq 2$ (Table V in Appendix A).

The picture changes when we also train under shot noise [Fig. 4(a) and (c)]. Trained and tested with 10^3 shots per image, AD stays within 1.5 percentage points of the noiseless accuracy at every depth, while the twirled channels lose up to 33 points at $L=4$. Part of this loss already arises in training. Evaluated without shot noise, the twirled models trained with shots remain clearly less accurate than those trained exactly, whereas AD loses almost nothing. After training with shots, the operational cost of AD stays close to its shot factor, whereas for the twirls it still grows by a factor of 3 to 5 per layer [Fig. 4(c)]. The reason is that training under shot noise moves the circuit toward expectation values whose spread exceeds the shot noise $1/\sqrt{N_s}$. Under AD, the smallest spread rises to the floor of Fig. 3, above the shot noise, while under the twirls it stays below it (Appendix A).

Finally, the learned rescaling is far cheaper than generic error mitigation. Probabilistic error cancellation [27] of the twirled channel would multiply the shots by $(1-p)^{-2}$ per insertion, or $(1-p)^{-72}\approx 10^{11}$ over the 36 insertions of the $L=4$ classifier, consistent with the exponential lower bounds on mitigation [28, 29]. The rescaling learned under shot noise, by comparison, needs at most 417 times the shots of the noiseless circuit. It restores, however, only the spread of the few expectation values that the task uses, not the noiseless state.

TABLE I. Dependence on the damping strength p before training, from the exact average over parameters: the depth L^* at which $(\sigma_{\rm AD}/\sigma_{\rm Pauli})^2$ first reaches 10, the floor $\sigma_{\rm br,\infty}$ of the AD spread and its last-layer part σ_∞ [Eq. (13)], and the number of shots $1/\sigma_{\rm br,\infty}^2$ that resolves the floor. The floor is estimated from 5000 ($n=4$) and 700 ($n=8$) random image pairs, with a sampling error of about 1%.

	$n=4$				$n=8$
p	L^*	$\sigma_{\rm br,\infty}$	σ_∞	$1/\sigma_{\rm br,\infty}^2$	L^*
0.001	1508	0.0017	0.0005	3.4×10^5	–
0.003	453	0.0031	0.0014	1.0×10^5	–
0.01	117	0.0067	0.0047	2.2×10^4	49
0.03	33	0.0155	0.0137	4.2×10^3	13
0.05	18	0.0235	0.0219	1.8×10^3	8
0.1	8	0.0408	0.0396	600	5
0.2	4	0.0647	0.0637	240	3
0.3	2	0.0757	0.0747	170	2

C. Dependence on the damping strength

So far we have worked at strong damping, $p=0.3$. We now turn to weaker damping, closer to current hardware, where both sides of the comparison change with p (Table I and Fig. 5). At weak damping, the twirled spread decays per layer as e^{-np}. This is because a scrambled Pauli string has about $3n/4$ non-identity sites, and each of the two insertions per layer shrinks such a site by a factor of about $1-2p/3$ on average. The floor of the AD spread, $\sigma_{\rm br,\infty}$, changes as well. Its square has two parts, $\sigma_{\rm br,\infty}^2 \approx \sigma_\infty^2 + b_{\rm br}\,p$. The first is proportional to p^2 [Eq. (13)]. The second comes from branchings in the roughly $1/(np)$ earlier layers that the damping does not yet suppress. It acts on the input dependence of those layers, which concentrates with width, so $b_{\rm br}$ falls by about an order of magnitude from $n=4$ to $n=8$.

To quantify when the two spreads separate, we define the separation depth L^* as the depth at which $(\sigma_{\rm AD}/\sigma_{\rm Pauli})^2$ first reaches 10 (Table I). There, the twirl needs ten times the shots of AD before training. Since the AD spread splits into the twirled part and a branching part $\sigma_{\rm br}$ (Appendix B),

$$\sigma_{\rm AD}^2 = \sigma_{\rm Pauli}^2 + \sigma_{\rm br}^2, \tag{15}$$

this happens when $\sigma_{\rm br} = 3\,\sigma_{\rm Pauli}$. With $\sigma_{\rm br}\approx\sigma_{\rm br,\infty}$ and

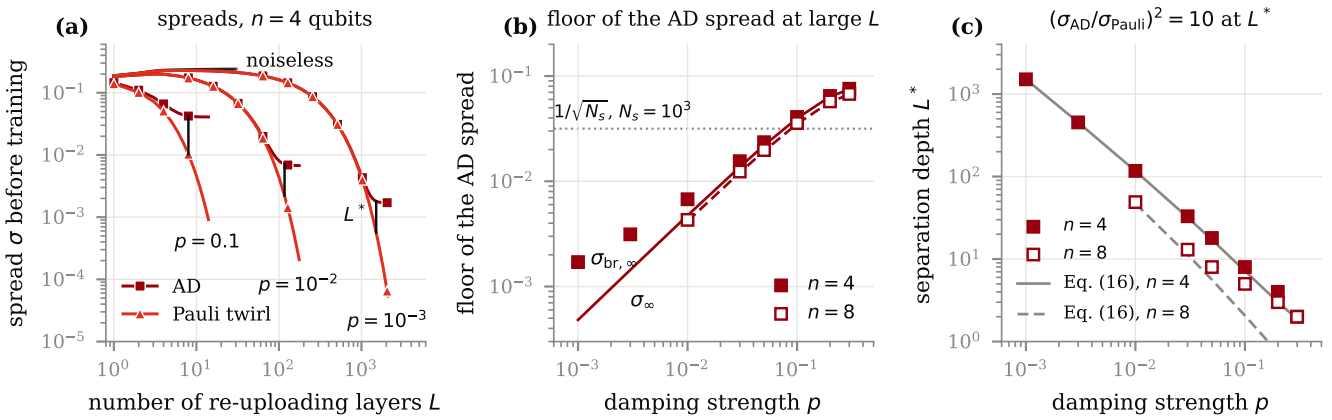


FIG. 5. Dependence of the separation on the damping strength, from the exact average over parameters before training (5000 image pairs at $n = 4$, 700 at $n = 8$). (a) Spread against depth at $n = 4$ for AD (squares) and the Pauli twirl (triangles) at three values of p, and for the noiseless circuit up to $L = 30$; vertical segments mark L^*. (b) Floor $\sigma_{\mathrm{br},\infty}$ of the AD spread (markers) and its last-layer part σ_∞ [Eq. (13), lines] for $n = 4$ (filled, solid) and $n = 8$ (hollow, dashed); the dotted line is the shot noise $1/\sqrt{N_s}$ at $N_s = 10^3$. (c) Separation depth L^* (markers) and Eq. (16) (gray lines). At $n = 8$ the exact average was computed for $p \geq 0.01$.

$\sigma_{\mathrm{Pauli}} \approx \sigma_{\mathrm{None}}\, e^{-npL}$, we obtain the estimate

$$L^* \approx \frac{1}{np} \ln \frac{3\,\sigma_{\mathrm{None}}}{\sigma_{\mathrm{br},\infty}}, \tag{16}$$

where σ_{None} is taken at large depth (0.24 at $n = 4$ and 0.062 at $n = 8$). At weak damping, this estimate agrees closely with the exact separation depth [Fig. 5(c)], and it grows roughly as $p^{-1} \ln(1/p)$. Interestingly, at fixed weak damping L^* falls faster than $1/n$ (Table I). This is because the noiseless spread in the logarithm concentrates with width faster than the floor does.

This has a clear consequence for real devices. On superconducting hardware, the damping is of order 10^{-3} per two-qubit gate, for gate times of 30 to 400 ns and T_1 of order 100 μs [30]. In our model, one of the two insertions per layer follows a ring of n CNOTs, so p per insertion corresponds to several gate durations. At $p = 10^{-3}$, the separation needs about 1500 layers at $n = 4$ (Table I). The difference in spread between AD and its twirls is therefore a property of strong damping or of very deep circuits.

Trained classifiers follow this picture (Table II). Well below the separation depth, at $p = 0.05$ and $L = 4$ ($L^* = 18$), AD and its twirls reach the same accuracy at nearly the same shot cost, as κ predicts. Closer to it, at $p = 0.1$ and $L = 4$ ($L^* = 8$), the twirls already need more shots than AD. At $L = L^* = 8$, the channels finally separate. The twirls then need about four times the shots of AD, in the same direction as the spread ratios before and after training. At this depth, AD is also 2.6 points more accurate than the Pauli twirl. It is not, however, more accurate than the best noiseless circuit (0.869 at $L = 2$ to 4), and with 10^3 shots per image it falls behind the noiseless circuit (Appendix A).

V. CONCLUSION

We have compared amplitude damping with its twirls, in particular with the Pauli twirl, which is damping at infinite temperature. In our classifier, the zero-temperature bias acts mainly through the spread of the expectation values. Under AD, the spread stops at a floor fixed at strong damping by the last layer, and at fixed T_1 this floor falls linearly in $1 - 2N$ as the temperature rises. Under the twirls, it keeps shrinking by a constant factor per layer at every width up to eight qubits.

Training an output scale removes most of the resulting differences in accuracy in exact simulation. What it removes reappears as a sampling cost. With 10^3 shots per image at $p = 0.3$, the classifier under AD stays within 1.5 percentage points of the noiseless accuracy at depths 1 to 4, while the twirled classifiers lose up to 33 points at four layers. The separation needs a depth that grows roughly as $(np)^{-1} \ln(1/p)$, about 1500 layers on four qubits at $p = 10^{-3}$, so it matters only at strong damping or in very deep circuits. This gives a simple reading of the reported degradation under twirling. The twirls shrink the scale at which the outputs must be read, more than they reduce what the model can learn.

Our results suggest a practice for comparing noise structures in variational models. A comparison should report the output scale, or train one, together with the shot budget it implies. Without the scale, a difference in accuracy can reflect the readout rather than the circuit. Without the shot budget, a trained scale hides the cost of the noise.

Several questions remain open. Our simulations are exact and small. We established the width dependence before training, up to eight qubits, but training beyond four qubits was out of reach of exact simulation, and our

TABLE II. Trained classifiers at weaker damping (MNIST, standardized readout, exact expectation values, eight initializations): test accuracy (mean ± SD), operational shot cost relative to the noiseless circuit at the same depth, shot factor κ, and the squared ratio of the AD and Pauli-twirl spreads before training, $(\sigma_{\mathrm{AD}}/\sigma_{\mathrm{Pauli}})^2$, from the exact average over parameters of Table I. The row $p = 0.3$ repeats Secs. III B and IV B.

		test accuracy				shot cost / noiseless			κ			before
p	L	None	AD	Pauli	Depol	AD	Pauli	Depol	AD	Pauli	Depol	training
0.3	4	$.869 \pm .005$	$.876 \pm .005$	$.870 \pm .007$	$.870 \pm .004$	967	1.7×10^4	1.4×10^4	50	1.6×10^4	1.1×10^4	1.3×10^3
0.05	4	$.869 \pm .005$	$.866 \pm .005$	$.866 \pm .007$	$.864 \pm .004$	4.7	4.8	4.1	4.6	4.7	4.4	1.06
0.1	4	$.869 \pm .005$	$.872 \pm .005$	$.863 \pm .004$	$.863 \pm .003$	13.1	21.5	14.9	15.6	21.2	17.5	1.69
0.1	8	$.831 \pm .012$	$.866 \pm .006$	$.840 \pm .009$	$.838 \pm .007$	115	466	495	62	484	459	16.8

shot model in training neglects the correlations between qubits. It also remains to be seen whether trained models follow the temperature dependence of the floor, and whether the results hold on hardware, where damping comes with dephasing.

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DATA AVAILABILITY

The code, the result files, and the scripts that reproduce every number and figure in this paper are available at https://github.com/Identity-Operator/zero-vs-infinite-T-damping and archived on Zenodo at doi:10.5281/zenodo.23053251 [31].

Appendix A: Statistical tests

We collect here the hypothesis tests behind the comparisons in the main text. Classifier results are given as the mean and standard deviation (SD) over eight initializations with fixed data. Because all runs share one finite test set, a difference between two conditions is uncertain through both the initializations and the test images. Accuracy differences therefore come with a 95% confidence interval (CI) from a two-level bootstrap [32] (2×10^4 resamples) that resamples both, with the initializations resampled jointly for the two conditions since they share their initial parameters. They also come with the P value of the paired t test across the eight initializations. To avoid confusion with the noise strength p, we denote P values by a capital P. The P values of Table III are Holm-adjusted over its 11 tests. Two remain below 0.05, both for AD, at $L = 3$ and 4.

TABLE III. Standardized-readout classifier on MNIST: difference from the noiseless circuit in test accuracy. P values from paired t tests over the eight initializations, Holm-adjusted over the 11 tests of the table.

L	channel	Δacc	95% CI	P
1	AD	+0.001	$[-0.003, +0.005]$	1
1	Pauli	−0.059	$[-0.178, +0.001]$	1
1	Depol	0	exact	–
2	AD	+0.003	$[-0.003, +0.009]$	1
2	Pauli	+0.004	$[-0.003, +0.010]$	1
2	Depol	+0.000	$[-0.004, +0.004]$	1
3	AD	+0.009	$[+0.003, +0.015]$	0.02
3	Pauli	+0.001	$[-0.007, +0.008]$	1
3	Depol	+0.000	$[-0.007, +0.007]$	1
4	AD	+0.008	$[+0.002, +0.013]$	0.004
4	Pauli	+0.001	$[-0.007, +0.009]$	1
4	Depol	+0.001	$[-0.006, +0.008]$	1

a. Classifier on Fashion-MNIST. These results are shown in Fig. 6. At $L = 4$, the differences from the noiseless circuit are +0.017, CI $[+0.007, +0.029]$, $P = 0.03$, for AD; +0.005, CI $[-0.004, +0.014]$, $P = 0.28$, for the Pauli twirl; and +0.009, CI $[-0.0001, +0.018]$, $P = 0.08$, for depolarizing noise (paired t, Holm-adjusted over the three channels).

b. Shot cost. At $L = 4$ with exact training, the operational shot cost of AD exceeds its shot factor 19-fold, with a 95% bootstrap interval over initializations of 12 to 30. The Pauli twirl needs 18 times the shots of AD on MNIST (10 to 29) and 36 times on Fashion-MNIST (17 to 85). Trained and tested with 10^3 shots, AD lies 0.7, 0.9, 1.2, and 1.5 points below the noiseless accuracy at $L = 1$ to 4, and at $L = 4$ it stays within 1.5 points also with 10^4 shots. Evaluated with exact expectation values, the twirled models trained with 10^3 shots are about 0.11 less accurate than those trained exactly, and AD about 0.01. At $L = 4$ and $N_s = 10^3$, the geometric mean of the smallest spread is 0.075 for AD, compared with 0.0065 under exact training and 0.24 for the noiseless circuit, and 0.005 and 0.003 for the Pauli and depolarizing chan-

TABLE IV. Test accuracy of the classifier on 3037 MNIST images (mean ± SD over eight initializations) against the number of re-uploading layers L. The raw readout is trained for 100 steps and, at $L = 4$, for 1000 steps; the standardized readout of Eq. (6) for 100 steps. †One of the eight runs diverged (test accuracy 0.382); the other seven lie between 0.860 and 0.865.

	raw readout					standardized readout			
	$L = 1$	$L = 2$	$L = 3$	$L = 4$	$L = 4$, 1000 steps	$L = 1$	$L = 2$	$L = 3$	$L = 4$
None	.861 ± .003	.866 ± .005	.865 ± .009	.863 ± .004	.862 ± .007	.860 ± .005	.869 ± .007	.868 ± .006	.869 ± .005
AD	.842 ± .010	.835 ± .032	.839 ± .005	.841 ± .004	.872 ± .003	.861 ± .003	.872 ± .003	.877 ± .004	.876 ± .005
Pauli	.847 ± .003	.817 ± .039	.787 ± .044	.715 ± .057	.837 ± .013	.802 ± .170†	.873 ± .005	.869 ± .009	.870 ± .007
Depol	.851 ± .003	.814 ± .041	.778 ± .034	.695 ± .041	.826 ± .012	.860 ± .005	.869 ± .006	.869 ± .010	.870 ± .004

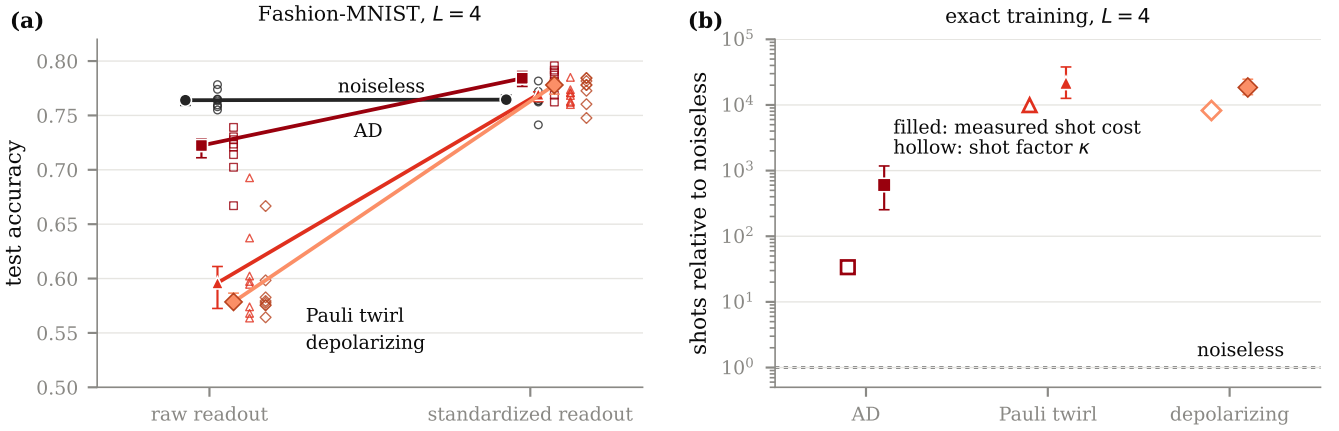


FIG. 6. Four-qubit classifier on Fashion-MNIST at $L = 4$ and $p = 0.3$. (a) Test accuracy with the raw and the standardized readout, as the median and interquartile range over eight initializations, with each run as a hollow marker. (b) Shot factor κ (hollow) and operational shot cost relative to the noiseless circuit (filled; geometric mean over initializations with 95% bootstrap intervals), both after exact training.

nels. Table V compares the exact-training costs with the noise gain G of Sec. IV B and with the shot factor.

c. Weaker damping. At $p = 0.1$ and $L = 8$ (Table II), AD minus the Pauli twirl in accuracy is +0.026, CI $[+0.017, +0.035]$. Evaluated with 10^3 shots per image, AD reaches 0.69 against 0.83 for the noiseless circuit.

Appendix B: Spread before training

The lines in Fig. 3 are exact averages, over parameters drawn uniformly from $[0, 2\pi)$, of Eq. (11), where the variance runs over the first 300 training images. We compute them in the Heisenberg picture from the second moments of the Pauli-string coefficients of Z_j, with the parameters shared between two images. Every uniformly random $R_Z R_Y R_Z$ block removes the cross moments between different strings because, averaged over its angles, the entries R_{bc} of its adjoint action have vanishing first moments and different columns are uncorrelated, $\mathbb{E}_\theta[R_{ab}R_{a'c}] = 0$ for $b \neq c$. Between two blocks the circuit acts site by site (channel, $R_X(a_j)$, channel), so the second moments of two images propagate through the elementwise product of the two composite site maps, followed by the entangling ring, which permutes the strings up to a sign. The second moments therefore obey a closed recursion, which is exact up to the sampling of image pairs: all pairs for $n \leq 5$, and 5000 random pairs for $n \geq 6$. Against brute-force averages of the density-matrix simulation, the recursion agrees within one to two standard errors in all 18 cells at $n = 2$ and 3 and $L = 1$ to 3. An independent density-matrix simulation at $n = 4$, with 200 parameter draws, agrees within two standard errors in all 12 cells for the noiseless circuit, AD, and the Pauli twirl at $L = 1$ to 4, and in all four cells for AD and the Pauli twirl at $p = 0.1$ and $L = 4$ and 8.

Under AD the adjoint channel maps Z to $c^2 Z + pI$. The next random block averages the cross moments between the I and Z branches to zero, which gives Eq. (15), with σ_{br} collecting the paths through at least one branching into the identity. The branching after the last entangler gives Eq. (12). Averaged over the final block, with $\mathbb{E}\, v_z^2 = \frac{1}{2}$, $\mathbb{E}\, v_y^2 = \frac{1}{4}$, and $\mathbb{E}\, v_z v_y = 0$, it yields Eq. (13). The agreement within 0.5% quoted in Sec. IV A uses all image pairs. Branchings in earlier layers are attenuated by the rest of the circuit. At weaker damping they are not, and they add the term $b_{\mathrm{br}}\, p$ of Sec. IV C. For uniformly distributed angles, $\sigma_\infty = 0.096$. The measured

TABLE V. Readout-weighted noise gain G of Sec. IV B compared with the operational shot cost of the classifiers trained with exact expectation values at $p = 0.3$. All entries are relative to the noiseless circuit and are geometric means over initializations. For $L \geq 2$, G matches the shot cost to within a factor of 1.6, whereas the shot factor κ underestimates the cost of AD by up to 19 times.

		shot cost / noiseless			G/G_{None}			κ		
dataset	L	AD	Pauli	Depol	AD	Pauli	Depol	AD	Pauli	Depol
MNIST	1	3.3	5.6	4.5	0.9	1.3	4.3	3.6	6.5	4.1
MNIST	2	17	125	60	14	125	58	7.9	79	52
MNIST	3	263	1.6×10^3	1.2×10^3	277	2.0×10^3	1.6×10^3	22	1.5×10^3	1.2×10^3
MNIST	4	967	1.7×10^4	1.4×10^4	983	2.4×10^4	1.5×10^4	50	1.6×10^4	1.1×10^4
Fashion-MNIST	4	603	2.2×10^4	1.8×10^4	400	1.6×10^4	1.2×10^4	34	9.9×10^3	8.2×10^3

points in Fig. 3 use five parameter draws, the same parameter vectors at every depth (those of depth L are the leading entries of those of depth 6), so their deviations are correlated across L. The distribution over the parameters is heavy-tailed, and five draws tend to fall below the average, as for the Pauli twirl at $n = 8$ and $L \geq 4$. An independent Pauli-path simulation reproduces these five draws to 2×10^{-12} and, over 120 further draws at $L = 4$, agrees with the exact average within one standard error.

In this exact average, the per-layer growth of the ratio of the AD and Pauli-twirl spreads, 3.3 at $n = 4$ (Sec. IV A), rises to 4.5 for $n \geq 6$, and that of its square from 11 to 20. At $n = 4$, the squared ratio is 2.1 at $L = 1$ and 1.3×10^3 at $L = 4$. The per-layer decay rate of the twirled spread agrees with np within 1% for $p \leq 0.05$ at $n = 4$ and 8. For $p \leq 0.03$ the coefficient b_{br} of Sec. IV C is (1.8–2.7) $\times 10^{-3}$ at $n = 4$ and (1.0–2.4) $\times 10^{-4}$ at $n = 8$, and Eq. (16) agrees with the exact separation depth within 1% for $p \leq 0.01$ and within one layer for larger p at $n = 4$, and within two layers for $p \leq 0.03$ at $n = 8$. At finite temperature, the branching part of the spread at $n = 4$ follows $1 - 2N$ within 0.4% for every $N \leq \frac{1}{2}$ and $L \leq 8$ at $p = 0.3$, and within 0.2% at $p = 0.1$.

Appendix C: Training details and numerical checks

a. Classifier training. To prepare the inputs, we standardize the pixel values, compute four principal components, and map each component linearly to an angle, sending its minimum to 0 and its maximum to 2π. The standardization, the principal components, and the extrema are fitted on 1000 training images and applied unchanged to all 3037 test images of the three digits. Training minimizes the softmax cross-entropy with Adam [33] at learning rate 0.05 for 100 full-batch steps unless stated otherwise, without weight decay. The circuit angles start uniformly in $[0, 2\pi)$, the readout weights from a zero-mean normal distribution with standard deviation 0.5, and the bias at zero. The density matrix is simulated exactly in PyTorch [34]. No regularizing constant is added to $\boldsymbol{\sigma}$ in the standardized readout. In one run, the spread of one z_j fell to 5×10^{-5}, and the training diverged (Table IV). The Fashion-MNIST classes are T-shirt, pullover, and coat, with 3000 test images. For reference, logistic regression, a k-nearest-neighbor classifier ($k = 15$), and a support-vector machine with a radial kernel, trained on the same angles, reach 0.856, 0.879, and 0.886 on MNIST and 0.733, 0.758, and 0.748 on Fashion-MNIST.

b. Shot model. For the finite-shot runs, each image is measured N_s times in the computational basis, and the estimate $\hat{z}_j$ of Eq. (9) is the mean of the outcomes ± 1 of qubit j. At test time we sample the outcomes from the exact distribution of the measured state, five times per model, and in every draw we re-estimate $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ from N_s shots of each training image, as one would on hardware. In training we replace each z_j by

$$\hat{z}_j = z_j + \sqrt{\frac{1 - z_j^2}{N_s}}\, \varepsilon_j, \tag{C1}$$

where ε_j is a standard normal random number, redrawn at every step and for every image, and compute $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ from the noisy $\hat{z}_j$; gradients flow through z_j. This Gaussian model has the exact shot variance of each z_j but neglects the correlations between qubits and the shot noise of parameter-shift gradients. We train with $N_s = 10^3$ at $L = 1$ to 4, and with $N_s = 10^2$ and 10^4 at $L = 4$. The operational shot cost is interpolated in $\log N_s$ on a grid from 10 to 10^8. It is undefined for three exactly trained models at $L = 1$, which we omit: the diverged Pauli run, and one noiseless and one depolarizing model whose accuracy stays more than 0.01 below its exact value up to 10^8 shots.

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